In Vitro Studies Indicate Promising Properties Of Solid Lipid Nanosuspensions For Oral Peptide Delivery

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Context and Objectives

Peptides are interesting therapeutic biomolecules with high specificity and low toxicity. However, they exhibit poor oral bioavailability: they are easily degraded by the proteolytic activity of the gastro-intestinal tract and their permeability across the intestinal epithelium is very low [1].

In this study, the potential of biocompatible and biodegradable lipid nanosuspensions to encapsulate and protect from enzymatic degradation a model peptide was evaluated.

In vitro efficiency of two systems was compared: Nanostructured Lipid Nanocarriers (NLC) and Solid Lipid Nanoparticles (SLN). Leuprolide (LEU) was used as model peptide.

- Develop nanosuspensions with increased Encapsulation Efficiency of a peptide in SLN and NLC by forming a Hydrophobic Ion Pair (HIP)
- Observe the nanosuspensions behavior in simulated intestinal fluids
- Evaluate the efficiency of the nanosuspensions to protect Leuprolide against proteolytic degradation
- Evaluate the interactions between nanosuspensions and Caco-2 cells and visualize the effect of mucus using Caco-2:HT29-MTX co-culture

Methods

Hydrophobic Ion Pairing

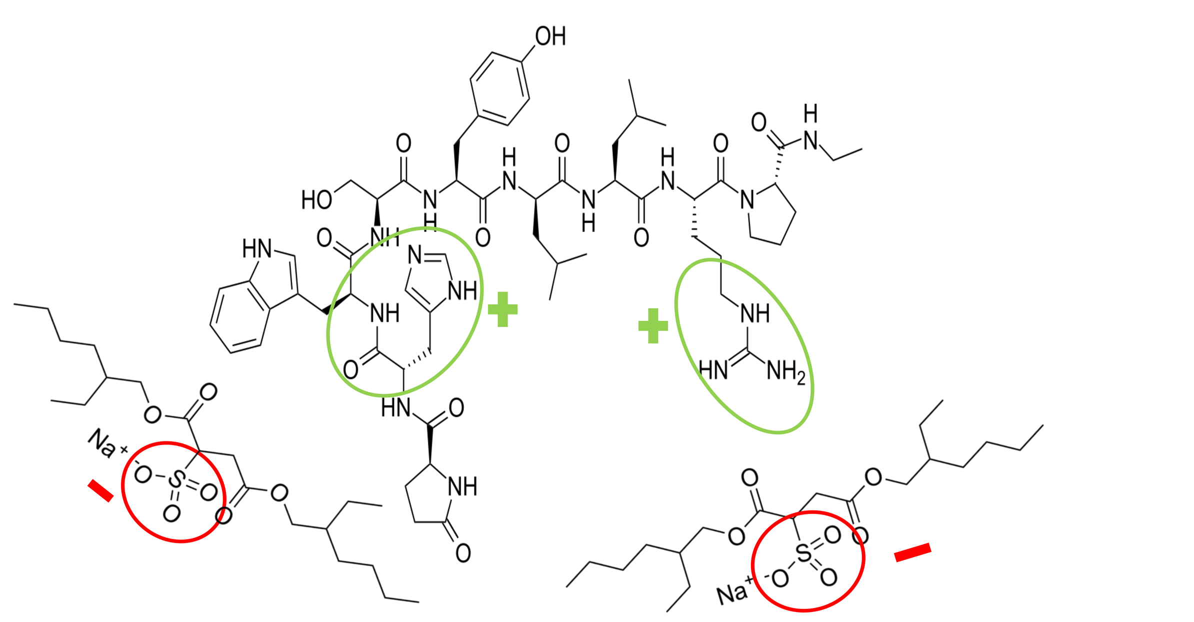


Fig. 1: Formation of a HIP by electrostatic interaction between Leuprolide and sodium docusate (molar ratio 1:2) [2]

$$\text{Precipitation Efficiency (PE, \%)} = 100 - \left(100X\frac{\text{Leuprolide concentration before HIP}}{\text{Leuprolide concentration after HIP}}\right)$$

Formulation by High Pressure Homogenization

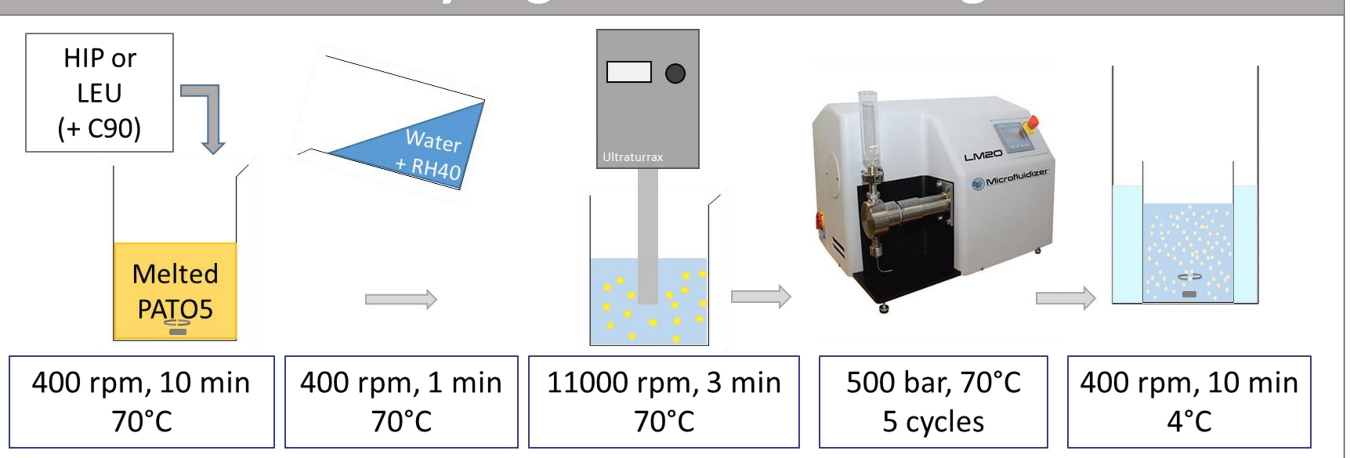


Fig. 2: Set up for the preparation of loaded SLN via HPH method

C90=Capryol™90 (propylene glycol monocaprylate), PATO5= Precirol®ATO5 (glyceryl distearate), RH40= Kolliphor®RH40 (polyoxyl casto oil)

$$\text{Encapsulation Efficiency (EE, \%)} = \frac{\text{Total amount of drug} - \text{unencapsulated amount of drug}}{\text{total amount of drug}} X100$$

In-vitro evaluation

Physico-chemical characterizations:
Particle size was determined by Diffraction Light Scattering (DLS) and confirmed by Cryo-TEM measurements.
PE and EE were measured by HPLC.

Interactions with simulated Gastro-intestinal Fluids:
LEU release was studied in FaSSIF-V2 over 6 hours.

Enzymatic degradation by trypsin:
Trypsin solution (0.3 mg/mL) was added to samples with a total LEU concentration of 100 µg/mL. Mixtures were placed at 37°C under stirring. Enzymatic activity was stopped at predetermined time points. Remaining LEU content was analyzed by HPLC.

Interactions of formulations with cell cultures:
Cytotoxicity was evaluated via MTT assay.
Interactions with Caco-2 were evaluated quantitatively by flow cytometry using SLN and NLC in which the HIP was co-encapsulated with DiD, a fluorescent dye. They were also evaluated qualitatively using a confocal microscope on Caco-2 and Caco-2:HT29-MTX (3:1) monolayers.

Results

Encapsulation Efficiency

Precipitation efficiency was **99.9%** (n=80).

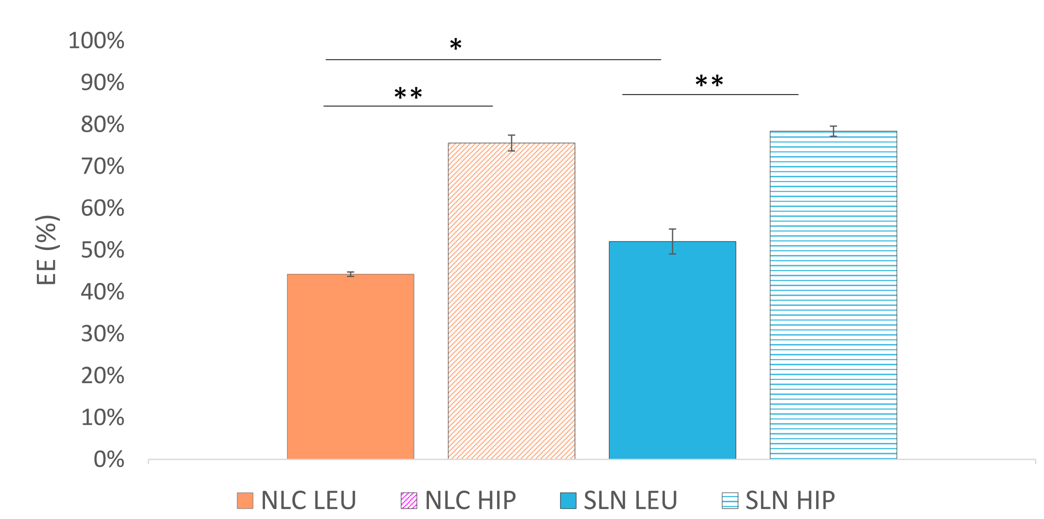


Fig. 3: HIP and LEU Encapsulation efficiency as a function of formulation type, (* p<0.05, ** p<0.01) (n=3)

Peptide release in FaSSIF-V2

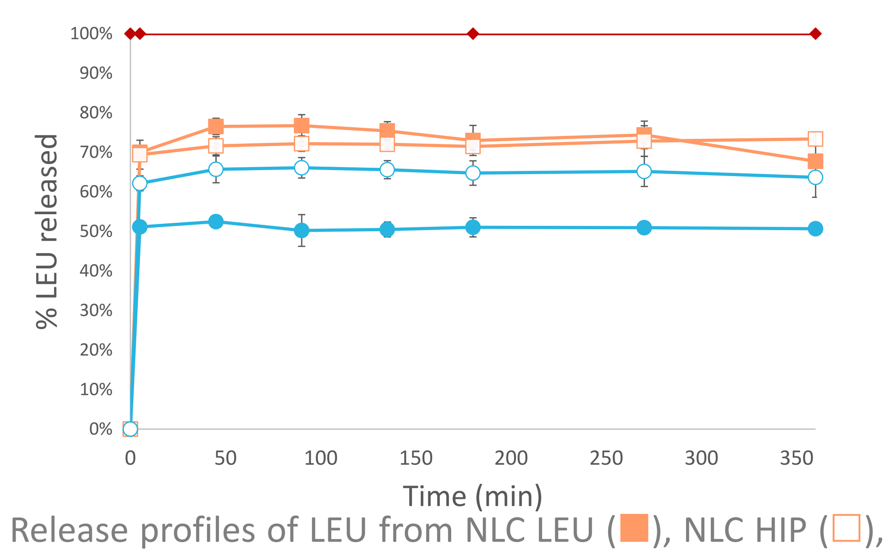


Fig. 5: Release profiles of LEU from NLC LEU (■), NLC HIP (□), SLN LEU (●) and SLN HIP (○) in FaSSIF-V2, compared to free LEU (◆) in FaSSIF.-V2

The platelet-like structure implies a large surface of exchange leading to an increased dissolution of LEU in FaSSIF-V2 [4].

Enzymatic degradation by trypsin

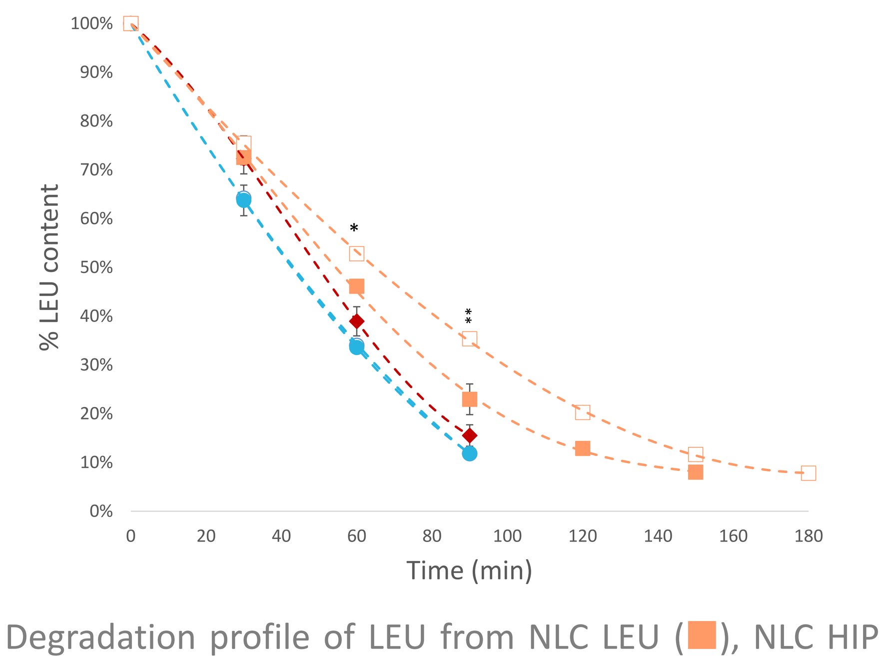


Fig 7: Degradation profile of LEU from NLC LEU (■), NLC HIP (□), SLN LEU (●) and SLN HIP (○) by trypsin, compared to free LEU (◆). Data expressed as mean ± SEM (n=3) (*p<0.05 and **p<0.01 when compared to suspensions dispersed in water)

Trypsin is not soluble in C90 [3]. Then, the addition of a C90 in NLC led to significant protection of LEU over trypsin degradation.

Physico-chemical characteristics

Table 1: Size distribution of NLC and SLN systems

	Blank NLC	NLC-LEU	NLC-HIP	Blank SLN	SLN-LEU	SLN-HIP
Z-average (nm)	114 ± 11	113 ± 1	125 ± 2	119 ± 4	124 ± 1	127 ± 1
PDI	0.2	0.2	0.2	0.2	0.2	0.2

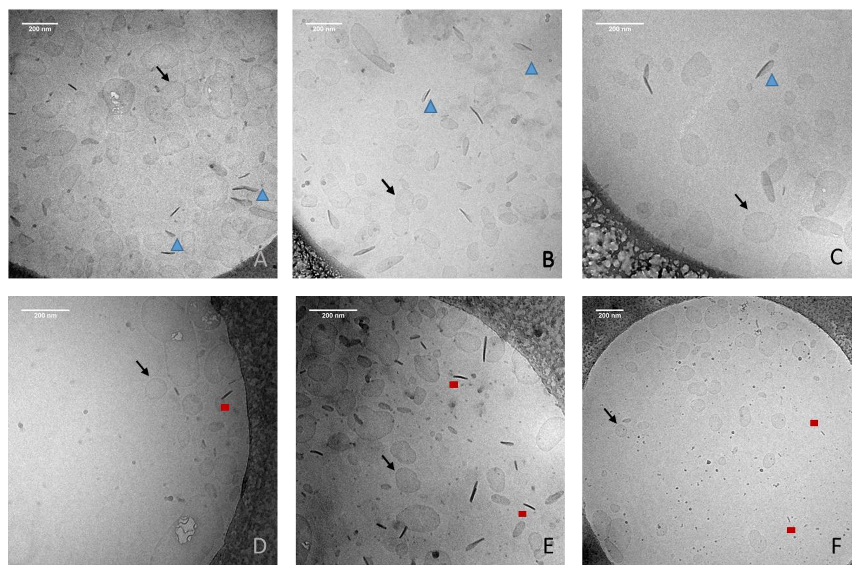


Fig. 4: Cryo-TEM images of blank NLC (A), NLC-LEU (B), NLC-HIP (C), blank SLN (D), SLN-LEU (E) and SLN-HIP (F). Black arrows point at top-view particles, the blue triangles indicate edge-on "nanospoon" platelets, red rectangles indicate edge-on platelet

Interaction with cells

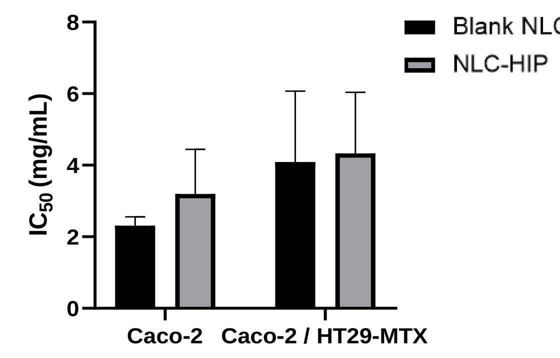
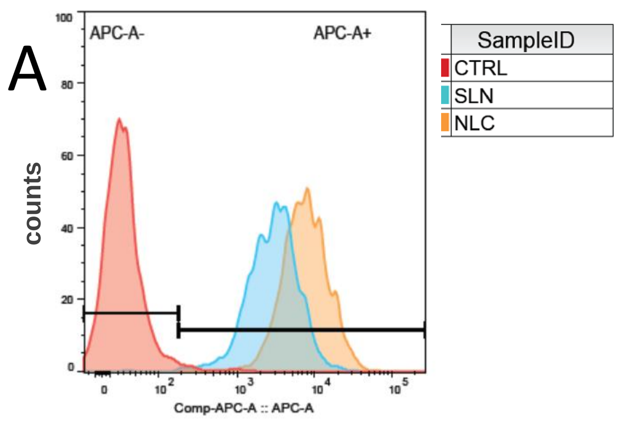


Fig. 8: IC_{50} values of blank NLC and HIP-loaded NLC in Caco-2 and Caco-2:HT29-MTX cells after incubation for 2 hours

Only NLC showed toxicity over Caco-2 or Caco-2:HT29-MTX co-cultures.



	% APC + /PE - cells	Mean Fluorescence intensity (a.u.)
Control	0.8 ± 0.1	32 ± 2
SLN	82.0 ± 4.6	1582 ± 438
NLC	98.8 ± 0.3	4558 ± 791

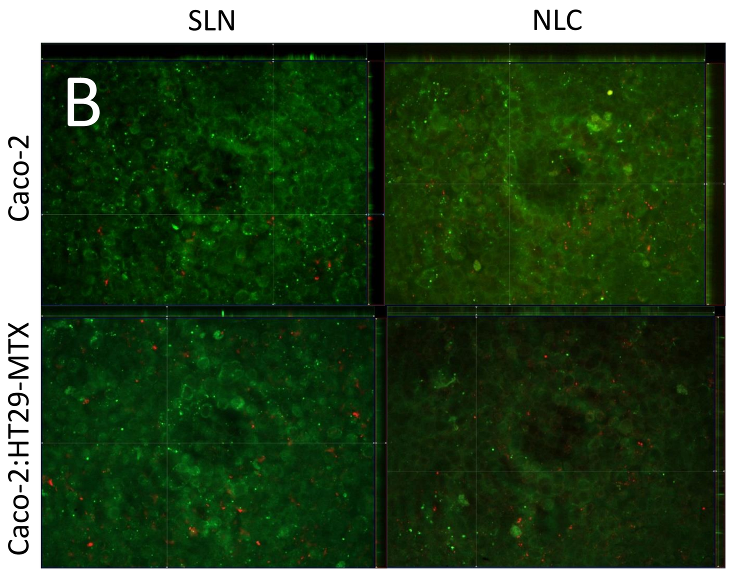


Fig. 9: (A) Flow cytometry quantitative interaction study of Caco-2 cells (N=3, n=10, data shown as mean ± SEM) (B) confocal microscopy images of SLN-DiD and NLC-DiD (red channels) interactions with Caco-2 monolayers stained with rhodamine-phalloidine (green channel), x25 (B)

The percentage of APC channel (DiD-NLC) positive cells and corresponding median fluorescence intensity clearly show the interaction of the nanoparticles with Caco-2 cells, with more internalization for NLC. Internalization of the particles is visible on both cell monolayers.

Conclusions

- Formation of HIP led to a significant increase in peptide encapsulation in solid lipid nanocarriers
- Inclusion of a liquid lipid fraction did not improve the EE
- NLC and SLN nanocarriers presented a similar size and morphology that lead to an important burst release in FaSSIF-V2 and that was not beneficial for peptide protection [4]
- Presence of a liquid fraction (C90) significantly impeded the enzymatic degradation of the peptide induced by trypsin
- Both NLC and SLN are able to penetrate Caco-2 with a higher internalization in the case of NLC. Nanosuspensions were internalized even in presence of mucus

References

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[2] Griesser, J., Hetényi, G., Moser, M., Demarne, F., Jannin, V., Bernkop-Schnürch, A., (2017). Hydrophobic ion pairing: Key to highly payloaded self-emulsifying peptide drug delivery systems. Int. J. Pharm. 520, 267–274.

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